

Burning
random trees

Luc Devroye
(McGill),
Austin Eide,
Paweł Prałat
(TMU)

Introduction

(Conditioned)
GW trees

Burning the
tree

Burning random trees

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September 2024

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Graph burning

Start with an “unburned” graph

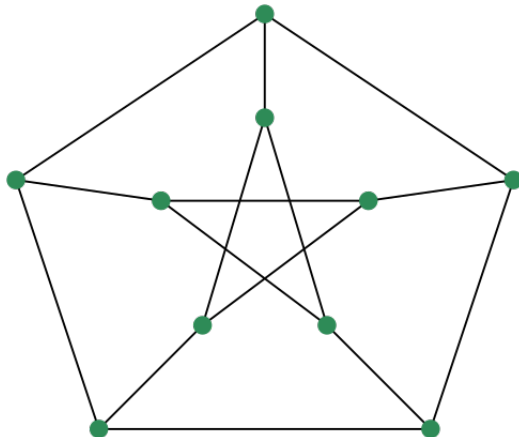
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Graph burning

At each step, “burn” one vertex. . .

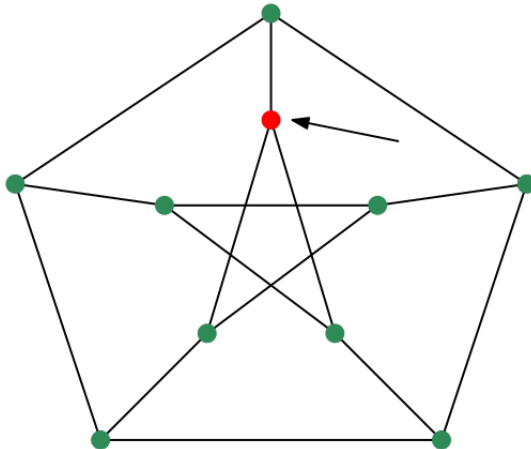
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Graph burning

... and fire spreads from burning vertices

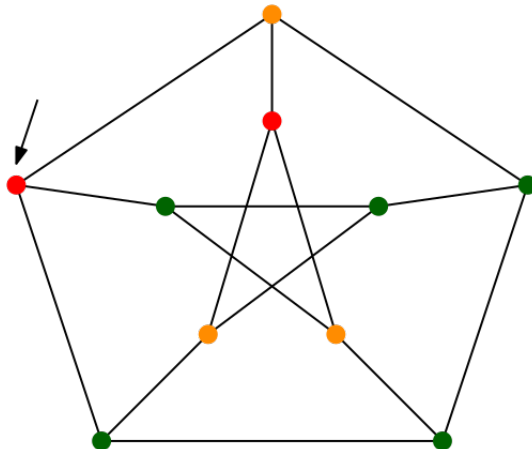
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Graph burning

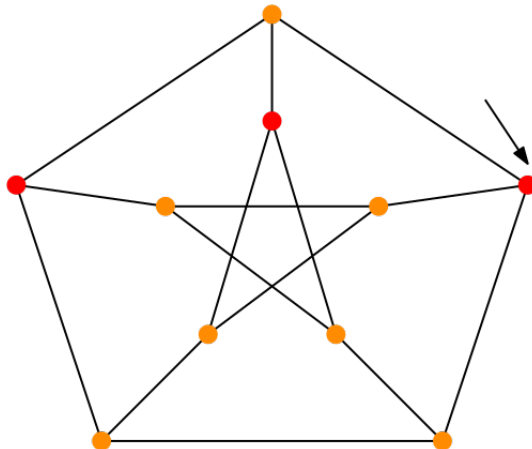
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Graph burning

This graph is burned after 3 rounds

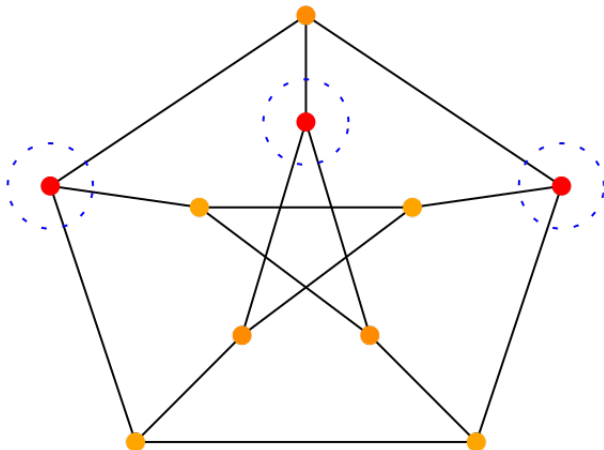
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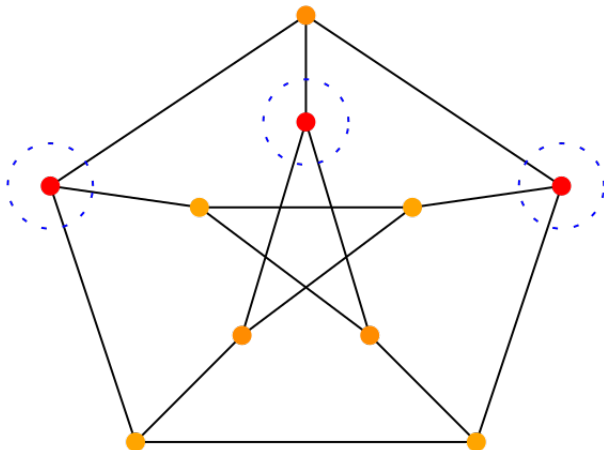
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Circled vertices are a *burning sequence*.

The burning number

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Definition

The *burning number* $b(G)$ is the length of a shortest burning sequence for G .

The burning number

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Claim: $b(\text{Petersen graph}) = 3$.

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The burning number

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Definition

The *burning number* $b(G)$ is the length of a shortest burning sequence for G .

Claim: $b(\text{Petersen graph}) = 3$. We demonstrated upper bound. For lower bound, note in 2 rounds we burn at most 5 vertices (have 10 total).

An alternative characterization

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Proposition

$(v_1, v_2, \dots, v_k)$ is a burning sequence iff

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Proposition

$(v_1, v_2, \dots, v_k)$ is a burning sequence iff

$$V(G) = N_{k-1}(v_1) \cup N_{k-2}(v_2) \cup \dots \cup N_0(v_k).$$

where $N_r(v) = r$ -neighborhood of v .

An alternative characterization

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where $N_r(v)$ = r -neighborhood of v .

So $b(G) \leq k$ iff G can be covered by k balls of increasing radii $0, 1, \dots, k-1$.

Example

The path

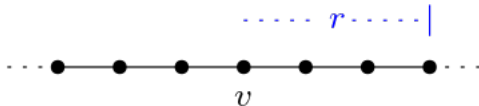
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On path, r -neighborhoods have size at most $2r + 1$.

Example

The path

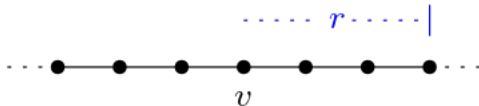
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On path, r -neighborhoods have size at most $2r + 1$.

If $b(P_n) \leq k$, then k satisfies

$$n \leq \sum_{r=0}^{k-1} (2r + 1) = k^2.$$

Example

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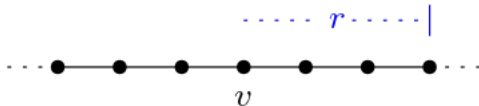
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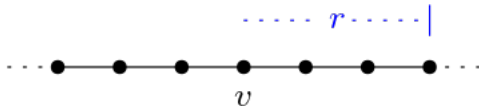
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i.e., $\lceil \sqrt{n} \rceil \leq k$. (In fact, $b(P_n) = \lceil \sqrt{n} \rceil$.)

The Burning Number Conjecture

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Conjecture (Bonato, Janssen, Roshanbin '16)

Every connected graph G on n vertices has $b(G) \leq \lceil \sqrt{n} \rceil$.

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Every connected graph G on n vertices has $b(G) \leq \lceil \sqrt{n} \rceil$.

Since $b(G) \leq b(T)$ for any spanning tree T of G , easy to restate as: **every tree T on n vertices has $b(T) \leq \lceil \sqrt{n} \rceil$.**

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Theorem (Norin & Turcotte '22)

If T is a tree on n vertices, then $b(T) \leq (1 + o(1))\sqrt{n}$. That is, the conjecture holds asymptotically.

The Burning Number Conjecture

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Best nonasymptotic bound is $\left\lceil \sqrt{\frac{4n}{3}} \right\rceil + 1$. (Bastide et. al. '21.)

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Rather than extreme case, we consider **typical** case.

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Rather than extreme case, we consider **typical** case.

Theorem (Devroye, E., Prałat '24)

Let T_n be a **random tree*** on n vertices. Then for any $\epsilon = o(1)$,

$$(\epsilon n)^{1/3} \leq b(T_n) \leq \left(\frac{n}{\epsilon}\right)^{1/3}$$

holds with probability tending to 1, i.e., w.h.p.

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*Galton Watson tree with offspring distribution ξ satisfying $E[\xi] = 1$, $\text{Var}[\xi] \in (0, \infty)$, conditioned to have n vertices.

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Let $\xi_1, \xi_2, \xi_3, \dots$ be i.i.d. copies of $\mathbb{Z}_{\geq 0}$ -valued r.v. ξ .

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Inductively define a rooted tree:

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- Root r has ξ_1 “offspring”

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- Root r has ξ_1 “offspring”
- First child has ξ_2 offspring, second has $\xi_3, \dots, \xi_1$ st has ξ_{ξ_1+1} .

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- etc.

Galton-Watson trees

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Galton-Watson trees

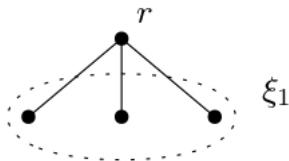
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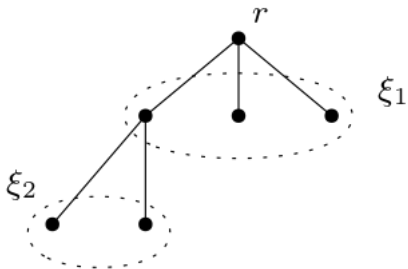
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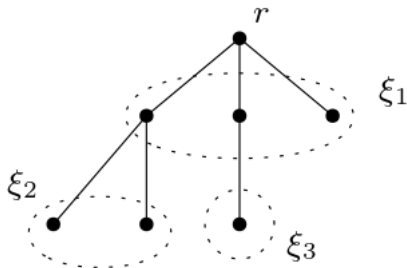
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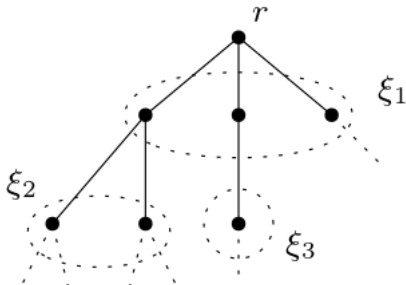
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Galton-Watson trees

Extinction and survival

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Theorem (Classical?)

Let $\mathcal{T}$ be a Galton-Watson tree with offspring distribution ξ , with $E[\xi] = \mu < \infty$ and $\text{Var}[\xi] = \sigma^2 \in (0, \infty)$.

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Extinction and survival

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- 1 *If $\mu \leq 1$, then $|\mathcal{T}| < \infty$ with probability 1, i.e., extinction.*

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Extinction and survival

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- 1 If $\mu \leq 1$, then $|\mathcal{T}| < \infty$ with probability 1, i.e., extinction.
- 2 If $\mu > 1$, then $|\mathcal{T}| = \infty$ with probability in $(0, 1)$.

We call $\mu < 1$ the *subcritical regime*, $\mu = 1$ *critical*, and $\mu > 1$ *supercritical*.

Critical Galton-Watson trees, conditioned on having n vertices

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Definition

Let $\mathcal{T}$ be a critical Galton-Watson tree with offspring distribution ξ . T_n is the tree $\mathcal{T}$ conditioned on the event $|\mathcal{T}| = n$.

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Let $\mathcal{T}$ be a critical Galton-Watson tree with offspring distribution ξ . T_n is the tree $\mathcal{T}$ conditioned on the event $|\mathcal{T}| = n$. (Some obvious arithmetic considerations notwithstanding.)

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- Has a limit object as $n \rightarrow \infty$

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- Random d -ary trees, plane trees, Catalan trees, Cayley trees, etc. realized as conditioned GW

Critical Galton-Watson trees, conditioned on having n vertices

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How to sample?

Pre-order degree sequences

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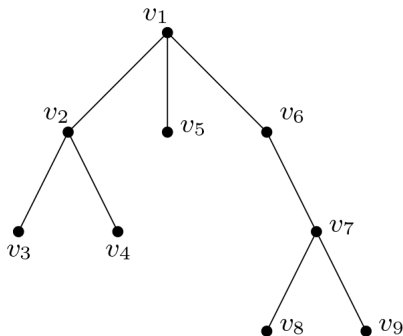
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Label vertices of τ in DFS order $(v_1, v_2, \dots, v_n)$:

Pre-order degree sequences

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Pre-order degree sequences

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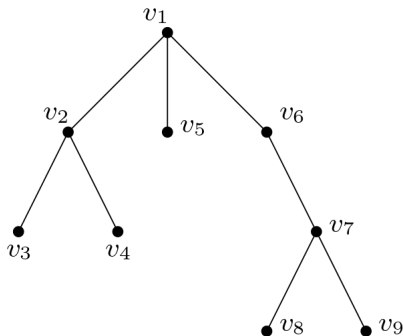
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Label vertices of τ in DFS order $(v_1, v_2, \dots, v_n)$:



Pre-order degree sequence:

$$(d_1, d_2, \dots, d_n) = (3, 2, 0, 0, 0, 1, 2, 0, 0)$$

Pre-order degree sequences

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Proposition

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$(d_1, d_2, \dots, d_n)$ a pre-order degree sequence for a tree $\iff$

$$\sum_{i=1}^j (d_i - 1) \geq 0 \quad \forall j \in \{1, \dots, n-1\}$$

and

Proposition

$(d_1, d_2, \dots, d_n)$ a pre-order degree sequence for a tree $\iff$

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and

$$\sum_{i=1}^n (d_i - 1) = -1$$

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and

$$\sum_{i=1}^n (d_i - 1) = -1$$

Proof.

$1 + \sum_{i=1}^j (d_i - 1)$ tracks # vertices in DFS queue at step j . $\square$

Pre-order degree sequences

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$(1, 2, 2, 0, 0, 1, 0)$



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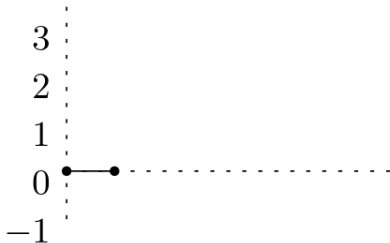
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$(1, 2, 2, 0, 0, 1, 0)$



Pre-order degree sequences

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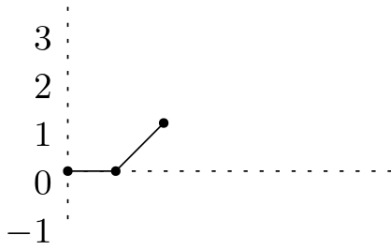
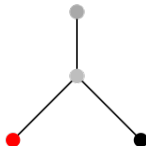
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$(1, 2, 2, 0, 0, 1, 0)$



Pre-order degree sequences

Burning
random trees

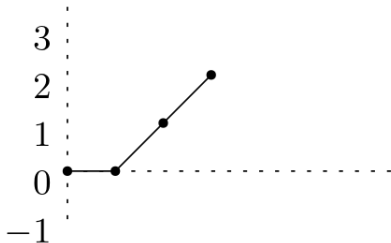
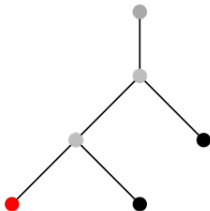
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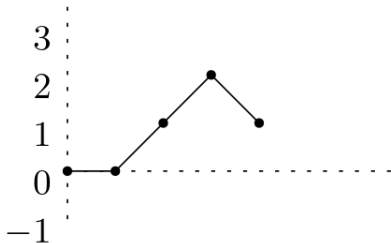
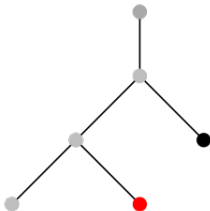
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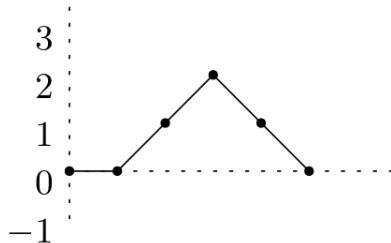
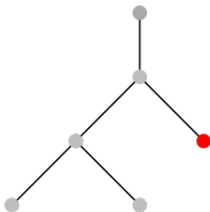
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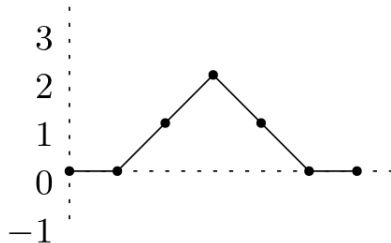
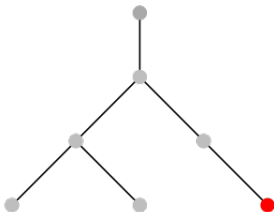
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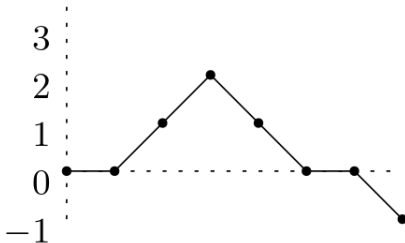
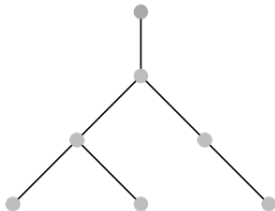
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Idea: sample T_n by sampling its degree sequence
 $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$.

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Idea: sample T_n by sampling its degree sequence
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Idea: sample T_n by sampling its degree sequence $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$. How to relate to $(\xi_1, \xi_2, \dots, \xi_n)$?

Crucial fact: if $\sum_{i=1}^n (d_i - 1) = -1$, then **exactly one** cyclic permutation of $(d_1, d_2, \dots, d_n)$ is the pre-order degree sequence of a tree.

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Relation: A random cyclic permutation of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$ is distributed as $(\xi_1, \xi_2, \dots, \xi_n)$, **conditioned on** $\sum_{i=1}^n (\xi_i - 1) = -1$,

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Idea: sample T_n by sampling its degree sequence $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$. How to relate to $(\xi_1, \xi_2, \dots, \xi_n)$?

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Relation: A random cyclic permutation of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$ is distributed as $(\xi_1, \xi_2, \dots, \xi_n)$, **conditioned on** $\sum_{i=1}^n (\xi_i - 1) = -1$, i.e., $\sum_{i=1}^n \xi_i = n - 1$.

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$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$ is the same as...

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$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$ is the same as...

... $(\xi_1, \xi_2, \dots, \xi_n)$ conditioned on $\sum_{i=1}^j (\xi_i - 1) \geq 0 \forall j$ and $\sum_{i=1}^n (\xi_i - 1) = -1$...

$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$ is the same as...

... $(\xi_1, \xi_2, \dots, \xi_n)$ conditioned on $\sum_{i=1}^j (\xi_i - 1) \geq 0 \forall j$ and $\sum_{i=1}^n (\xi_i - 1) = -1$...

... and taking a random cyclic permutation removes the first condition.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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Fact

If G can be covered with $2k$ balls of radius $2k$, then $b(G) \leq 4k$.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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Fact

If G can be covered with $2k$ balls of radius $2k$, then $b(G) \leq 4k$.

Since we only care about order of magnitude, suffices to cover with k balls of radius k , with $k = \left(\frac{n}{\epsilon}\right)^{1/3}$

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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To cover a rooted tree τ with balls of radius $2k$:

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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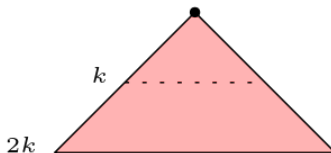
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To cover a rooted tree τ with balls of radius $2k$:



Put ball of radius $2k$ at the root.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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Choose parameter $j \in \{0, 1, \dots, k-1\}$. Examine level j .

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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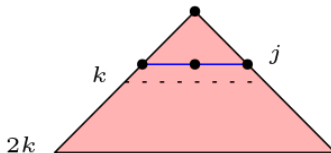
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If v on level j has a subtree of height at least k , put ball of radius $2k$ on v .

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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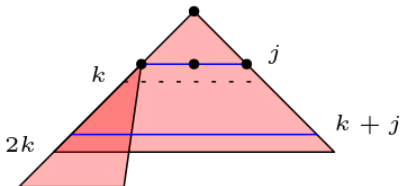
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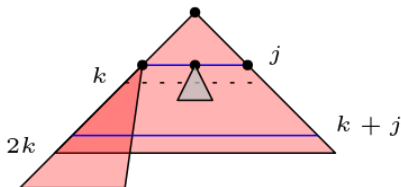
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If not, skip!

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

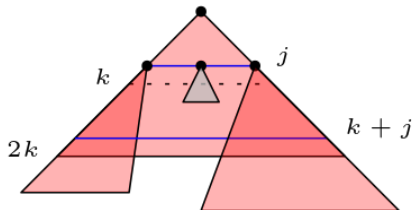
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Upper bound

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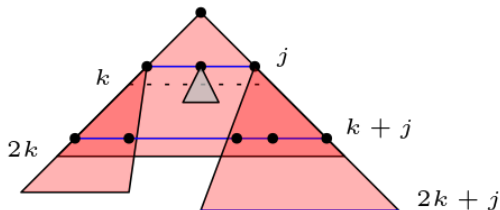
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Repeat for $k + j, 2k + j, \dots$



Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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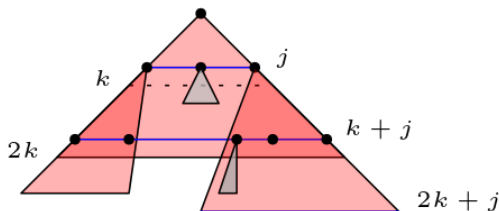
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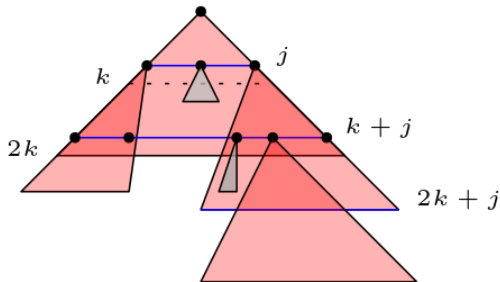
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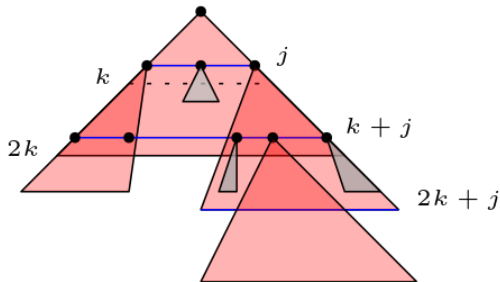


Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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treeRepeat for $k + j$, $2k + j$, ...

We never miss a vertex!

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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If there are at most $2k - 1$ vertices with subtrees of height $\geq k$
on levels $j, k + j, 2k + j, \dots$ for **SOME** j

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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If there are at most $2k - 1$ vertices with subtrees of height $\geq k$ on levels $j, k + j, 2k + j, \dots$ for **SOME** j then we cover τ with $2k$ balls of radius $2k$, and $b(\tau) \leq 4k$.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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Theorem

Let $\tau = T_n$. Let u be a uniformly random vertex, and τ_u the full subtree rooted at u .

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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Theorem

Let $\tau = T_n$. Let u be a uniformly random vertex, and τ_u the full subtree rooted at u . Then uniformly in $0 \leq k \leq O(\sqrt{n})$

$$\Pr(\text{height}(\tau_u) \geq k) = O\left(\frac{1}{k}\right).$$

In particular, we expect $O\left(\frac{n}{k}\right)$ subtrees of height $\geq k$.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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W.h.p., $\leq \frac{n}{\epsilon k}$ deep subtrees. . .

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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W.h.p., $\leq \frac{n}{\epsilon k}$ deep subtrees. $\dots k$ options for the partition
parameter $j \in \{0, 1, \dots, k-1\} \dots$

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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W.h.p., $\leq \frac{n}{\epsilon k}$ deep subtrees. $\dots k$ options for the partition
parameter $j \in \{0, 1, \dots, k-1\} \dots$ some choice of j has at most
a $\frac{1}{k}$ fraction of all deep subtrees, i.e., $\leq \frac{n}{\epsilon k^2}$.

Upper bound

$$b(T_n) \leq (n/\epsilon)^{1/3}$$

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W.h.p., $\leq \frac{n}{\epsilon k}$ deep subtrees. $\dots k$ options for the partition parameter $j \in \{0, 1, \dots, k-1\}$. $\dots$ some choice of j has at most a $\frac{1}{k}$ fraction of all deep subtrees, i.e., $\leq \frac{n}{\epsilon k^2}$.

Now, choose k so that $\frac{n}{\epsilon k^2} \leq 2k - 1$, i.e., $\left(\frac{n}{\epsilon}\right)^{1/3} \leq k$.

Theorem

Let $\tau = T_n$. Let u be a uniformly random vertex, and τ_u the full subtree rooted at u . Then uniformly in $0 \leq k \leq O(\sqrt{n})$

$$\Pr(\text{height}(\tau_u) \geq k) = O\left(\frac{1}{k}\right).$$

Upper bound

Proof idea

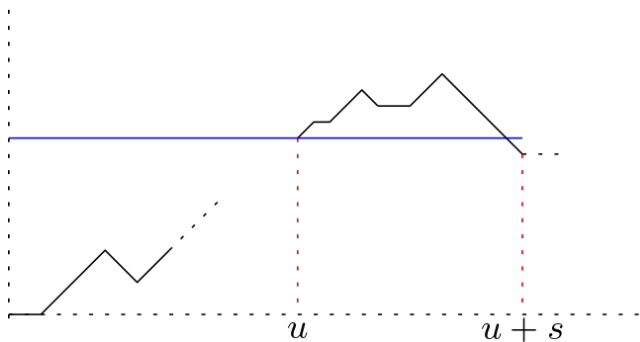
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Let u be a random vertex of T_n :



Lattice path representation of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})$.

Upper bound

Proof idea

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Choosing $u \iff$ choosing cyclic shift of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})!$

Upper bound

Proof idea

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Choosing $u \iff$ choosing cyclic shift of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})!$

$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)}) \mapsto$

Upper bound

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Choosing $u \iff$ choosing cyclic shift of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})!$

$$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)}) \mapsto (\xi_u^{(n)}, \dots, \xi_n^{(n)}, \xi_1^{(n)}, \dots, \xi_{u-1}^{(n)})$$

Upper bound

Proof idea

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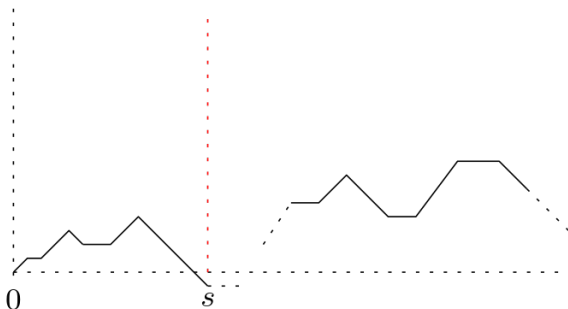
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Choosing $u \iff$ choosing cyclic shift of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})!$

$$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)}) \mapsto (\xi_u^{(n)}, \dots, \xi_n^{(n)}, \xi_1^{(n)}, \dots, \xi_{u-1}^{(n)})$$



Upper bound

Proof idea

Burning
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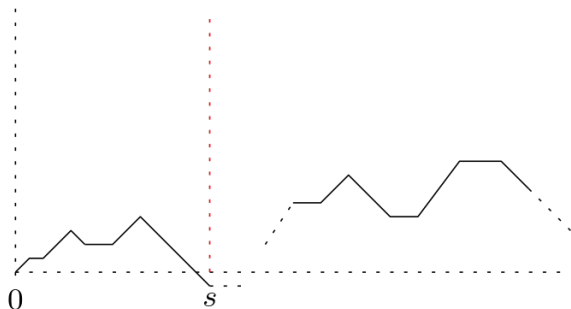
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Choosing $u \iff$ choosing cyclic shift of $(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)})!$

$$(\xi_1^{(n)}, \xi_2^{(n)}, \dots, \xi_n^{(n)}) \mapsto (\xi_u^{(n)}, \dots, \xi_n^{(n)}, \xi_1^{(n)}, \dots, \xi_{u-1}^{(n)})$$



Now distributed as $(\xi_1, \xi_2, \dots, \xi_n)$ cond. on $\sum_{i=1}^n \xi_i = n - 1$.

Upper bound

Proof idea

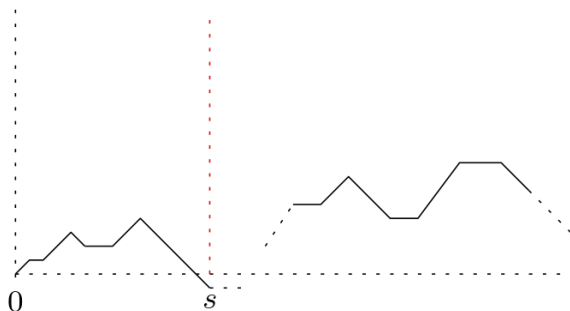
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Initial s -segment is T_s ... and final $n-s$ -segment is $(\xi_1, \xi_2, \dots, \xi_{n-s})$ conditioned on $\sum_{i=1}^{n-s} \xi_i = n-s$.

Lower Bound

$$b(T_n) \geq (n/\epsilon)^{1/3}$$

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Show **no covering** with k balls of radii $\leq k$ exists.

Lower Bound

$$b(T_n) \geq (n/\epsilon)^{1/3}$$

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Show **no covering** with k balls of radii $\leq k$ exists.

Theorem (Devroye, Janson)

Uniformly in $i \geq 0$, we expect $O(n \cdot i)$ pairs v, w in T_n with $d(v, w) = i$.

Lower Bound

$$b(T_n) \geq (n/\epsilon)^{1/3}$$

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Theorem (Devroye, Janson)

Uniformly in $i \geq 0$, we expect $O(n \cdot i)$ pairs v, w in T_n with $d(v, w) = i$.

At most $\frac{nk^2}{\epsilon}$ pairs at distance $\leq 2k$ in T_n w.h.p.

Lower Bound

$$b(T_n) \geq (n/\epsilon)^{1/3}$$

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Questions?